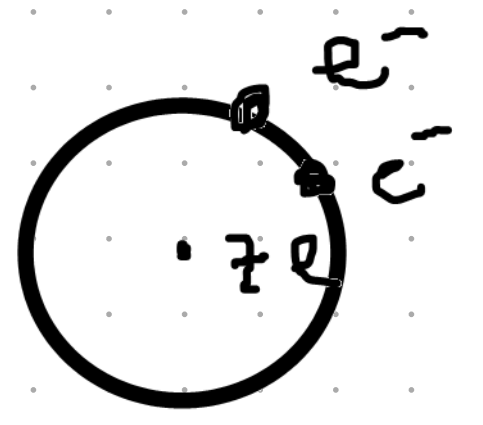


He Atom - Varyation



$$H = \frac{p_1^2}{2m} + \frac{p_2^2}{2m} - \frac{ze^2}{r_1} - \frac{ze^2}{r_2} + \frac{e^2}{|r_1 - r_2|}$$

$$\psi_{100} = \sqrt{\frac{4}{\pi}} \left(\frac{z}{a_0} \right)^{3/2} e^{-zr/a_0}$$

$$H\psi = E\psi \Rightarrow \langle H \rangle = E$$

$$\langle \psi | H | \psi \rangle = E \langle \psi | \psi \rangle$$

$$\int d^3r_1 \int d^3r_2 \psi_{100}^*(r_1) \psi_{100}^*(r_2) H \psi_{100}(r_1) \psi_{100}(r_2)$$

$$\int d^3r_1 \int d^3r_2 \psi_{100}^*(r_1) \psi_{100}^*(r_2) \left(\frac{p_1^2}{2m} + \frac{p_2^2}{2m} - \frac{ze^2}{r_1} - \frac{ze^2}{r_2} + \frac{e^2}{|r_1 - r_2|} \right) \psi_{100}(r_1) \psi_{100}(r_2)$$

$$\rightarrow r_1 \Rightarrow \int d^3r_1 \psi_{100}^*(r_1) \left(\frac{p_1^2}{2m} - \frac{ze^2}{r_1} \right) \psi_{100}(r_1) \quad \psi_{100} = \sqrt{\frac{4}{\pi}} \left(\frac{z}{a_0} \right)^{3/2} e^{-zr/a_0} = N(z) e^{-zr/a_0}$$

$$E = -\frac{1}{2} mc^2 \left(\frac{z\alpha}{n} \right)^2 \Rightarrow \text{vary } a_0 \text{ i.e. } z \Rightarrow -\frac{1}{2} mc^2 \left(\frac{z^* \alpha}{n} \right)^2 = E \Rightarrow \int d^3r_1 \psi_{100}^*(r_1) \left(\frac{p_1^2}{2m} - \frac{ze^2}{r_1} \right) \psi_{100}(r_1)$$

$$E + \int d^3r_1 \psi_{100}^*(r_1) \left(\frac{ze^2}{r_1} - \frac{ze^2}{r_1} \right) \psi_{100}(r_1) =$$

$$E + e^2 (z^* - z) \int d^3r_1 \underbrace{\psi_{100}^*(r_1) \psi_{100}(r_1)}_{z^*/a_0} = E + e^2 (z^* - z) \frac{z^*}{a_0}$$

$$\int d^3r_2 \int d^3r_1 \underbrace{\psi_{100}^*(r_1) \psi_{100}^*(r_2) \left(\frac{e^2}{|r_1 - r_2|} \right) \psi_{100}(r_1) \psi_{100}(r_2)}_{\frac{5z^*}{8a_0}}$$

$\times 2e^-$

$$\frac{5z^*}{8a_0}$$

$$2E + 2(z^* - z) \frac{z^*}{a_0} + \frac{5z^*}{8a_0} \Rightarrow E = -\frac{1}{2} mc^2 \left(\frac{z\alpha}{n} \right)^2$$

$$a_0 = \frac{1}{mc^2 \alpha^2}$$

$$2E + 2(z - z^*) z^* (mc^2 \alpha^2) + \frac{5}{8} z^* mc^2 \alpha^2$$

$$2 \left(-\frac{1}{2} mc^2 \left(\frac{z^* \alpha}{n} \right)^2 \right) + 2(z - z^*) z^* (mc^2 \alpha^2) + \frac{5}{8} z^* mc^2 \alpha^2$$

$$-\frac{1}{2} mc^2 \alpha^2 \left(\frac{2(z^*)^2}{n^2} + 4(z - z^*) z^* - \frac{5}{4} z^* \right) =$$

$$\Rightarrow -\frac{1}{2} mc^2 \alpha^2 \left(2z^{*2} + 4zz^* - 4z^{*2} - \frac{5}{4} z^* \right)$$

$$\frac{dE(z)}{dz^*} = -\frac{1}{2} mc^2 \alpha^2 \left(-2z^{*2} + 4zz^* + \frac{5}{4} z^* \right) = -4z^* + 4z - \frac{5}{4} = 0$$

$$-4z^* = +\frac{5}{4} - 4z$$

$$z^* = \frac{5}{16} + z \Rightarrow E_n = -77.38 \text{ eV}$$

Böhm 16 7. amanda Boşmsta Rüt. Te

$$H\Psi = E\Psi \Rightarrow H|\Psi\rangle = E|\Psi\rangle \quad H = H_0 + \lambda H_1 + \lambda^2 H_2 \quad |\Psi\rangle = |\phi_n\rangle + \sum_{n \neq k} c_{nk} \lambda^1 |\phi_k\rangle + \sum_{n \neq k} c_{nk} \lambda^2 |\phi_k\rangle$$

$$E_n = E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2$$

$$\Rightarrow (H_0 + \lambda H_1 + \lambda^2 H_2) \left(|\phi_n\rangle + \sum_{n \neq k} c_{nk} \lambda^1 |\phi_k\rangle + \sum_{n \neq k} c_{nk} \lambda^2 |\phi_k\rangle \right) = (E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2) \left(|\phi_n\rangle + \sum_{n \neq k} c_{nk} \lambda^1 |\phi_k\rangle + \sum_{n \neq k} c_{nk} \lambda^2 |\phi_k\rangle \right)$$

$$\underline{H_0|\phi_n\rangle} + \sum_{n \neq k} \underline{c_{nk} \lambda H_0|\phi_k\rangle} + \sum_{n \neq k} \underline{c_{nk} \lambda^2 H_0|\phi_k\rangle} + \lambda \underline{H_1|\phi_n\rangle} + \sum_{n \neq k} \underline{c_{nk} \lambda^2 H_1|\phi_k\rangle} = \underline{E_n^0|\phi_n\rangle} + \sum_{n \neq k} \underline{c_{nk} \lambda E_n^0|\phi_k\rangle} + \sum_{n \neq k} \underline{c_{nk} \lambda^2 E_n^0|\phi_k\rangle} + \lambda \underline{E_n^1|\phi_n\rangle} + \sum_{n \neq k} \underline{c_{nk} \lambda E_n^1|\phi_k\rangle} + \lambda^2 \underline{E_n^2|\phi_n\rangle}$$

$$\lambda^0 \Rightarrow H_0|\phi_n\rangle = E_n^0|\phi_n\rangle$$

$$\lambda^1 \Rightarrow \sum_{n \neq k} c_{nk} H_0|\phi_k\rangle + H_1|\phi_n\rangle = \sum_{n \neq k} c_{nk} E_n^0|\phi_k\rangle + E_n^1|\phi_n\rangle$$

$$\lambda^2 \Rightarrow \sum_{n \neq k} c_{nk} H_0|\phi_k\rangle + \sum_{n \neq k} c_{nk} H_1|\phi_k\rangle = \sum_{n \neq k} c_{nk} E_n^0|\phi_k\rangle + \sum_{n \neq k} c_{nk} E_n^1|\phi_k\rangle + E_n^2|\phi_n\rangle$$

$$\lambda^1 \Rightarrow \sum_{n \neq k} c_{nk} H_0|\phi_k\rangle + H_1|\phi_n\rangle = \sum_{n \neq k} c_{nk} E_n^0|\phi_k\rangle + E_n^1|\phi_n\rangle$$

$$\langle \phi_n | \Rightarrow \sum_{n \neq k} c_{nk} \langle \phi_n | H_0 | \phi_k \rangle + \langle \phi_n | H_1 | \phi_n \rangle = \sum_{n \neq k} c_{nk} E_n^0 \langle \phi_n | \phi_k \rangle + E_n^1 \langle \phi_n | \phi_n \rangle$$

$$\rightarrow \underline{\langle \phi_n | H_1 | \phi_n \rangle = E_n^1 \langle \phi_n | \phi_n \rangle}$$

$$\langle \phi_m | \Rightarrow \sum_{n \neq k} c_{nk} \langle \phi_m | H_0 | \phi_k \rangle + \langle \phi_m | H_1 | \phi_n \rangle = \sum_{n \neq k} c_{nk} E_n^0 \langle \phi_m | \phi_k \rangle + E_n^1 \langle \phi_m | \phi_n \rangle$$

$$\sum_{n \neq k} c_{nk} \langle \phi_m | H_0 | \phi_k \rangle + \langle \phi_m | H_1 | \phi_n \rangle = \sum_{n \neq k} c_{nk} E_n^0 \langle \phi_m | \phi_k \rangle + E_n^1 \langle \phi_m | \phi_n \rangle$$

$$\sum_{n \neq k} c_{nk} E_k^0 \langle \phi_m | \phi_k \rangle = \sum_{n \neq k} c_{nk} E_n^0 \langle \phi_m | \phi_k \rangle$$

$$\langle \phi_m | H_1 | \phi_n \rangle = \sum_{n \neq k} c_{nk} (E_n^0 - E_k^0) \langle \phi_m | \phi_k \rangle$$

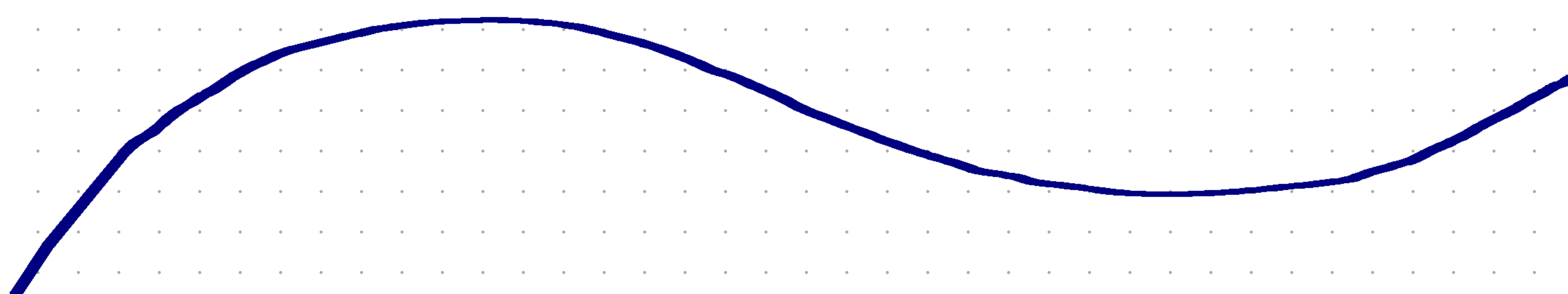
$$\rightarrow \Rightarrow \underline{\frac{\langle \phi_m | H_1 | \phi_n \rangle}{E_n^0 - E_m^0} = \sum_{m \neq n} c_{mn}}$$

$$\sum_{n \neq k} c_{nk} H_0|\phi_k\rangle + \sum_{n \neq k} c_{nk} H_1|\phi_k\rangle = \sum_{n \neq k} c_{nk} E_n^0|\phi_k\rangle + \sum_{n \neq k} c_{nk} E_n^1|\phi_k\rangle + E_n^2|\phi_n\rangle$$

$$\langle \phi_k | \Rightarrow \sum_{n \neq k} c_{nk} \langle \phi_k | H_0 | \phi_k \rangle + \sum_{n \neq k} c_{nk} \langle \phi_k | H_1 | \phi_k \rangle = \sum_{n \neq k} c_{nk} E_n^0 \langle \phi_k | \phi_k \rangle + \sum_{n \neq k} c_{nk} E_n^1 \langle \phi_k | \phi_k \rangle + E_n^2 \langle \phi_k | \phi_n \rangle$$

$$\langle \phi_n | \Rightarrow \sum_{n \neq k} c_{nk} \langle \phi_n | H_0 | \phi_k \rangle + \sum_{n \neq k} c_{nk} \langle \phi_n | H_1 | \phi_k \rangle = \sum_{n \neq k} c_{nk} E_n^0 \langle \phi_n | \phi_k \rangle + \sum_{n \neq k} c_{nk} E_n^1 \langle \phi_n | \phi_k \rangle + E_n^2 \langle \phi_n | \phi_n \rangle$$

$$\underline{\frac{\langle \phi_m | H_1 | \phi_n \rangle}{E_n^0 - E_m^0} = \sum_{m \neq n} c_{mn}} \Rightarrow \sum_{m \neq n} \frac{|\langle \phi_m | H_1 | \phi_n \rangle|^2}{E_n^0 - E_m^0} = E_n^2$$



Problems

1. Consider the hydrogen atom, and assume that the proton, instead of being a point-source of the Coulomb field, is a uniformly charged sphere of radius R , so that the Coulomb potential is now modified to

$$V(r) = -\frac{3e^2}{2R^3} \left(R^2 - \frac{1}{3} r^2 \right) \quad r < R (\ll a_0)$$

$$= -\frac{e^2}{r} \quad r > R$$

Calculate the energy shift for the $n = 1, l = 0$ state, and for the $n = 2$ states, caused by this modification, using the wave functions given in (12-25).

* Actually a simple barrier penetration calculation of the type carried out in Chapter 5 shows that the time scale is more like 10^{1000} lifetimes of the universe, for fairly reasonable fields!

268

Quantum Physics

$$V(r) = -\frac{3e^2}{R} + \frac{e^2}{2R^3} r^2 \quad r < R$$

$$= -\frac{e^2}{r} \quad r > R$$

$$H_1 = -\frac{3e^2}{R} + \frac{e^2}{2R^3} r^2 + \frac{e^2}{r} \quad r < R$$

$$= -\frac{e^2}{r} \quad r > R$$

$$(H_0 + \lambda H_1 + \lambda^2 H_2) \psi = (E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2) \psi$$

$$\lambda H_1 \Rightarrow \langle \phi_n | H_1 | \phi_n \rangle = E_n^1 \langle \phi_n | \phi_n \rangle \Rightarrow$$

$$\langle \psi_{100} | H_1 | \psi_{100} \rangle = E_1^1 \langle \psi_{100} | \psi_{100} \rangle$$

$$\psi_{100} = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$$

$$\int d\mathbf{r} \psi_{100}^* H_1 \psi_{100} =$$

$$\int d\Omega \int r^2 dr \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0} \left(-\frac{3e^2}{R} + \frac{e^2 r^2}{2R^3} + \frac{e^2}{r} \right) \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0} =$$

$$\frac{4\pi}{\pi a_0^3} \int r^2 dr e^{-2r/a_0} \left(-\frac{3e^2}{R} + \frac{e^2 r^2}{2R^3} + \frac{e^2}{r} \right)$$

$$\Rightarrow \frac{4e^2}{a_0^3} \left\{ \frac{1}{R} \int_0^R r^2 dr e^{-2r/a_0} + \frac{1}{2R^3} \int_0^R r^4 dr e^{-2r/a_0} + \int_0^\infty r dr e^{-2r/a_0} \right\} \quad \Gamma(x) = \int_0^\infty e^{-t} t^{x-1} dt = (x-1)!$$

$$\Gamma(x) = \int_0^\infty e^{-t} t^{x-1} dt \quad 2r/a_0 = t \Rightarrow r = \frac{a_0 t}{2} \Rightarrow dr = \frac{a_0}{2} dt$$

$$(i) \int \frac{a_0^3 t^2}{4} \frac{a_0}{2} dt e^{-t} = \frac{a_0^3}{8} \int \frac{t^2}{2} e^{-t} dt = \Gamma(3) = 2! = 2 = \frac{a_0^3}{4R}$$

$$(ii) \frac{1}{2R^3} \int \frac{a_0^4}{16} + 4 \left(\frac{a_0}{2} \right) dt e^{-t} = \frac{1}{2R^3} \frac{a_0^5}{32} \int t^4 e^{-t} dt \Rightarrow \frac{24 a_0^5}{64 R^3} = \frac{3 a_0^5}{8 R^3} = (iii)$$

$$\Gamma(5) = 4! = 24$$

$$(iii) \int r dr e^{-r/a_0} = \int_0^\infty \frac{a_0 t}{2} \left(\frac{a_0}{2} \right) dt e^{-t} = \frac{a_0^2}{4} \int t e^{-t} dt = \frac{a_0^2}{4} = (iv)$$

$$\Gamma(2) = 1!$$

$$\frac{4e^2}{a_0^3} \left(\frac{a_0^3}{4R} + \frac{3 a_0^5}{8 R^3} + \frac{a_0^2}{4} \right) = \left(\frac{e^2}{R} + \frac{3}{2} \frac{a_0^2 e^2}{R^3} + \frac{e^2}{a_0} \right)$$

2. Calculate the energy shift in the ground state of the one-dimensional harmonic oscillator, when the perturbation

$$V = \lambda x^4 \quad V = \lambda x^2$$

is added to

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

3. Consider a square well in one dimension. If the edges of the well are rounded off as shown in the figure, what is the change in the ground state energy?

Choose your rounding-off parametrization such that $\int_{-\infty}^{\infty} V(x) dx$ remains constant.

$$A^\dagger = \sqrt{\frac{m\omega}{2\hbar}} x - \frac{i p}{\sqrt{2m\omega}}$$

$$A = \sqrt{\frac{m\omega}{2\hbar}} x + \frac{i p}{\sqrt{2m\omega}}$$

$$A^\dagger + A = \sqrt{\frac{2m\omega}{\hbar}} x \Rightarrow (A^\dagger + A) \sqrt{\frac{\hbar}{2m\omega}} = x$$

$$x^2 = \frac{\hbar}{2m\omega} (A^\dagger + A)^2 \Rightarrow \frac{\hbar}{2m\omega} (A^{\dagger 2} + A^2 + A^\dagger A + A A^\dagger)$$

$$\Rightarrow \frac{\hbar}{2m\omega} (A^{\dagger 2} + A^2 + 1 + 2A A^\dagger)$$

$$H = H_0 + \lambda H_1 + \lambda^2 H_2 \Rightarrow E_n' = \lambda \langle \psi | H_1 | \psi \rangle$$

$$A^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle \quad A |n\rangle = \sqrt{n} |n-1\rangle$$

$$\lambda \frac{\hbar}{2m\omega} \langle n | A^{\dagger 2} + A^2 + A^\dagger A + A A^\dagger | n \rangle \Rightarrow$$

$$\lambda \frac{\hbar}{2m\omega} \left\{ \begin{aligned} &\langle n | A^\dagger \sqrt{n+1} | n+1 \rangle + \langle n | A \sqrt{n} | n-1 \rangle + \langle n | A^\dagger \sqrt{n} | n-1 \rangle + \langle n | A \sqrt{n+1} | n+1 \rangle \\ &\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ &\langle n | \sqrt{n+1} \sqrt{n+2} | n+2 \rangle \quad \langle n | \sqrt{n} \sqrt{n-1} | n-2 \rangle \quad \sqrt{n} \langle n | \sqrt{n} | n \rangle \quad \sqrt{n+1} \langle n | \sqrt{n+1} | n \rangle \\ &\quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \downarrow \quad \downarrow \\ &\quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad n \langle n | n \rangle = n \quad + \quad n+1 \langle n+1 | n+1 \rangle = n+1 \end{aligned} \right\}$$

$$\Rightarrow \lambda \frac{\hbar}{2m\omega} (2n+1) \Rightarrow$$

$$E_n = \hbar \omega \left(n + \frac{1}{2} \right) \Rightarrow \lambda \frac{\hbar}{m\omega} \left(n + \frac{1}{2} \right)$$

$$\Rightarrow TDE = \hbar \omega \left(n + \frac{1}{2} \right) \Rightarrow \left(\hbar \omega + \frac{\lambda \hbar}{m\omega} \right) \left(n + \frac{1}{2} \right) \Rightarrow \hbar \omega_0 \left(n + \frac{1}{2} \right)$$

$\hbar \omega_0$

$$H = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & c \\ 0 & 0 & -2 \end{pmatrix} + \begin{pmatrix} 0 & c & 0 \\ c & 0 & 0 \\ 0 & 0 & c \end{pmatrix}$$

a) H in 3x3 matrix

b) Pert. too small to solve

$$a) H = \begin{pmatrix} 1 & c & 0 \\ c & 3 & 0 \\ 0 & 0 & -2+c \end{pmatrix} \quad \text{eigenvalues} \quad \det(Mat - \lambda I)$$

$$\text{Eigenvalue} = \begin{vmatrix} 1-\lambda & c & 0 \\ c & 3-\lambda & 0 \\ 0 & 0 & -2-\lambda \end{vmatrix} = \begin{vmatrix} 1-\lambda & c & 0 \\ c & 3-\lambda & 0 \\ 0 & 0 & -2-\lambda \end{vmatrix} = \begin{vmatrix} 1-\lambda & c & 0 \\ c & 3-\lambda & 0 \\ 0 & 0 & -2-\lambda \end{vmatrix} = 0$$

$$(1) (c-2-\lambda) \cdot \begin{vmatrix} 1-\lambda & c \\ c & 3-\lambda \end{vmatrix} = 0 \Rightarrow (c-2-\lambda) \{ (1-\lambda)(3-\lambda) - c^2 \} = 0$$

$$(c-2-\lambda) \{ \lambda^2 - 4\lambda + 3 - c^2 \} = 0$$

$$\lambda_1, \lambda_2, \lambda_3 \Rightarrow \lambda_1 = 2-c$$

$$\lambda_2, \lambda_3 = ? \quad \Delta = ? \quad b^2 - 4ac = 16 - 4(3-c^2)$$

$$16 - 12 + 4c^2 = 4 + 4c^2 \Rightarrow 4(1+c^2)$$

$$\text{roots} = \frac{-b \pm \sqrt{\Delta}}{2a} \Rightarrow$$

$$\frac{4 \pm \sqrt{4(1+c^2)}}{2} = \frac{4 \pm 2\sqrt{1+c^2}}{2} = 2 \pm \sqrt{1+c^2} = \lambda_2, \lambda_3$$

$$b) H_0 + \lambda H_1 \quad H = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & c \\ 0 & 0 & -2 \end{pmatrix} + \begin{pmatrix} 0 & c & 0 \\ c & 0 & 0 \\ 0 & 0 & c \end{pmatrix}$$

$$H_0 \Rightarrow E_0^{ii} \Rightarrow E_0^{11} \quad E_0^{22} \quad E_0^{33}$$

$\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 $1 \quad \quad 3 \quad \quad -2$

$$H_1 \Rightarrow E_1^{ii} \Rightarrow E_1^{11} \quad E_1^{22} \quad E_1^{33} \quad E_1 = \langle \phi_i | H_1 | \phi_i \rangle$$

$\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 $0 \quad \quad 0 \quad \quad c$

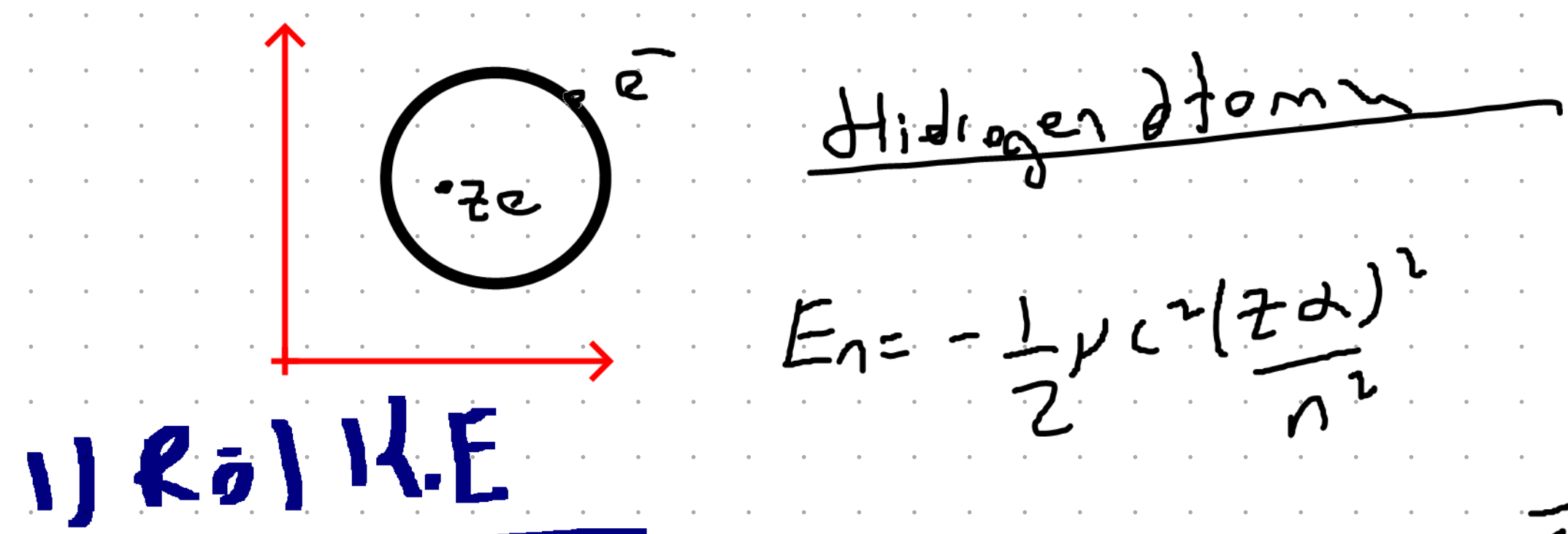
$$E_1^{22} = ? \quad \frac{|\langle \phi_n | H_1 | \phi_m \rangle|^2}{E_n^0 - E_m^0} \Rightarrow E_2^{ii} = \sum_{i \neq j} \frac{\langle i | H_1 | j \rangle \langle j | H_1 | i \rangle}{E_n^0 - E_m^0} + 0 \quad j=2 \quad i=3$$

$$\frac{c^2}{E_1^0 - E_3^0} + 0 = \frac{c^2}{1-3} = -\frac{c^2}{2}$$

$$E_2^{ii} \quad j=3 \Rightarrow \sum_{i \neq j} \frac{\langle i | H_1 | j \rangle \langle j | H_1 | i \rangle}{E_1^0 - E_3^0} + \frac{\langle 2 | H_1 | 3 \rangle \langle 3 | H_1 | 2 \rangle}{E_2^0 - E_3^0}$$

$\underbrace{\hspace{10em}}_0 \quad \quad \underbrace{\hspace{10em}}_0$

Bölüm 18



$$E_n = -\frac{1}{2} \mu c^2 \left(\frac{Z\alpha}{n} \right)^2$$

1) R0) K.E

$$P^\mu = \left(\frac{E}{c}, \vec{p} \right) \quad P_\mu = \left(\frac{E}{c}, -\vec{p} \right) \Rightarrow P^\mu P_\mu = \frac{E^2}{c^2} - \vec{p}^2 \Rightarrow \bar{p} = mc \Rightarrow p^2 = m^2 c^2$$

$$\Rightarrow m^2 c^2 = \frac{E^2}{c^2} - p^2 \Rightarrow \boxed{m^2 c^4 + p^2 c^2 = E^2}$$

$$\boxed{m^2 c^4 + p^2 c^2 = E^2} \Rightarrow E = \pm \sqrt{m^2 c^4 + p^2 c^2} \Rightarrow$$

$$\Rightarrow E = + (m^2 c^4 + p^2 c^2)^{1/2} \quad E = K + mc^2$$

$$\Rightarrow K = E - mc^2 \Rightarrow K = (m^2 c^4 + p^2 c^2)^{1/2} - mc^2$$

$$\Rightarrow (m^2 c^4 \{ 1 + \frac{p^2 c^2}{m^2 c^4} \})^{1/2} - mc^2 = K \Rightarrow$$

$$K = mc^2 (1 + \frac{p^2 c^2}{m^2 c^4})^{1/2} - mc^2$$

$$K = mc^2 \left\{ \left(1 + \frac{p^2 c^2}{m^2 c^4} \right)^{1/2} - 1 \right\}$$

(1+x)^n seri açılımı

$$\Rightarrow (1+x)^n = 1 + n \frac{x}{1} + n(n-1) \frac{x^2}{2!} \Rightarrow$$

$$\Rightarrow \left(1 + \frac{p^2 c^2}{m^2 c^4} \right) = 1 + \frac{1}{2} \cdot \frac{p^2 c^2}{m^2 c^4} + \frac{1}{2} \left(\frac{1}{2} \right) \frac{(p^2 c^2)^2}{2(m^2 c^4)^2}$$

$$\Rightarrow 1 + \frac{1}{2} \frac{p^2 c^2}{m^2 c^4} - \frac{1}{8} \left(\frac{p^2 c^2}{m^2 c^4} \right)^2$$

$$1 + \frac{1}{2} \frac{p^2 c^2}{m^2 c^4} - \frac{1}{8} \left(\frac{p^2 c^2}{m^2 c^4} \right)^2 \approx \left(1 + \frac{p^2 c^2}{m^2 c^4} \right)^{1/2} \Rightarrow mc^2 \left\{ 1 + \frac{1}{2} \frac{p^2 c^2}{m^2 c^4} - \frac{1}{8} \left(\frac{p^2 c^2}{m^2 c^4} \right)^2 - 1 \right\}$$

$$\Rightarrow mc^2 \left(\frac{1}{2} \frac{p^2 c^2}{m^2 c^4} - \frac{1}{8} \frac{p^4 c^4}{m^4 c^8} \right) \Rightarrow$$

$$\Rightarrow \frac{1}{2} \frac{p^2}{m} - \frac{1}{8} \frac{p^4}{m^3 c^2} \Rightarrow H_0 = \frac{p^2}{2m} + V(r) \quad \frac{-Ze^2}{r} \Rightarrow H_0 + \frac{Ze^2}{r} = \frac{p^2}{2m}$$

$$\frac{p^2}{2m} - \left(\frac{p^2}{2m} \right)^2 \frac{1}{2mc^2} \Rightarrow H_0 + \frac{Ze^2}{2m} - \underbrace{\left(\frac{p^2}{2m} \right)^2 \left(\frac{1}{2mc^2} \right)}_{H_1}$$

$$H_0 = \frac{p^2}{2m} - \frac{Ze^2}{r} \Rightarrow \frac{p^2}{2m} = H_0 + \frac{Ze^2}{r}$$

$$H_1 = \left(\frac{p^2}{2m} \right)^2 \left(\frac{1}{2mc^2} \right) \Rightarrow H_1 = \left(H_0 + \frac{Ze^2}{r} \right)^2 \left(\frac{1}{2mc^2} \right)$$

$$\langle H_1 \rangle = ? \Rightarrow \langle \psi_{nlm} | H_1 | \psi_{nlm} \rangle \Rightarrow$$

$$\langle \psi_{nlm} | \frac{1}{2mc^2} (H_0 + \frac{Ze^2}{r})^2 | \psi_{nlm} \rangle$$

$$\Rightarrow \frac{1}{2mc^2} \langle \psi_{nlm} | (H_0 + \frac{Ze^2}{r}) (H_0 + \frac{Ze^2}{r}) | \psi_{nlm} \rangle$$

$$\frac{1}{2mc^2} \left\{ \langle \psi_{nlm} | H_0^2 | \psi_{nlm} \rangle + \langle \psi_{nlm} | H_0 \frac{Ze^2}{r} | \psi_{nlm} \rangle + \langle \psi_{nlm} | \frac{Ze^2}{r} H_0 | \psi_{nlm} \rangle + \langle \psi_{nlm} | \frac{Ze^2}{r^2} | \psi_{nlm} \rangle \right\}$$

$$\frac{1}{2mc^2} \langle \psi_{nlm} | (H_0 + \frac{ze^2}{r}) (H_0 + \frac{ze^2}{r}) | \psi_{nlm} \rangle$$

$$\frac{1}{2mc^2} \left\{ \langle \psi_{nlm} | H_0^2 | \psi_{nlm} \rangle + \langle \psi_{nlm} | H_0 \frac{ze^2}{r} | \psi_{nlm} \rangle + \langle \psi_{nlm} | \frac{ze^2}{r} H_0 | \psi_{nlm} \rangle + \langle \psi_{nlm} | \frac{z^2 e^4}{r^2} | \psi_{nlm} \rangle \right\}$$

$$\frac{1}{2mc^2} \left\{ E_n^0{}^2 + E_n^0 \underbrace{ze^2 \langle \psi_{nlm} | \frac{1}{r} | \psi_{nlm} \rangle}_{\frac{z}{a_0 n^2}} + E_n^0 ze^2 \underbrace{\langle \psi_{nlm} | \frac{1}{r} | \psi_{nlm} \rangle}_{\frac{z^2}{a_0^2 n^3 (l + \frac{1}{2})}} + \underbrace{z^2 \langle \psi_{nlm} | \frac{1}{r^2} | \psi_{nlm} \rangle}_{\frac{z^2}{a_0^2 n^3 (l + \frac{1}{2})}} \right\}$$

$$\frac{z}{a_0 n^2}$$

$$\frac{z^2}{a_0^2 n^3 (l + \frac{1}{2})}$$

$$\frac{1}{2mc^2} \left\{ E_n^0{}^2 + 2 E_n^0 \frac{ze^2}{a_0 n^2} + \frac{z^2 (ze^4)^2}{a_0^2 n^3 (l + \frac{1}{2})} \right\}$$

$$\left\{ \frac{-1}{2} mc^2 \left(\frac{za}{n} \right)^2 \right\}^2$$

$$\Rightarrow \frac{1}{2mc^2} \left\{ \frac{1}{4} \frac{m^2 c^4 (za)^4}{n^4} - \frac{mc^4 (za)^2 ze^2}{a_0 n^4} + \frac{z^2 (ze^4)^2}{a_0^2 n^3 (l + \frac{1}{2})} \right\}$$

$$\Rightarrow \frac{1}{8} \frac{mc^2 (za)^4}{n^4} - \frac{1}{2} \frac{(za)^2 ze^2}{a_0 n^4} + \frac{z^2 (ze^4)^2}{2mc^2 a_0^2 n^3 (l + \frac{1}{2})}$$

$$- \frac{1}{2} mc^2 (za)^2 \left[-\frac{1}{4} \frac{(za)^2}{n^4} + \frac{ze^2 a^2}{mc^2 a_0 n^4} - \frac{ze^2 a^2}{n^4 a_0^2 n^3 (l + \frac{1}{2})} \right]$$

$$\Rightarrow \frac{1}{2} mc^2 (za)^2 \left[\frac{3}{4} \frac{ze^2 a^2}{n^4} - \frac{ze^2 a^2}{n^3 (l + \frac{1}{2})} \right]$$

$$\underbrace{\hspace{10em}}_{H_1}$$

$$\boxed{\alpha = l, \quad mc^2 a_0 = 1}$$

